

Communicating in time and space - How to overcome incompatible frames of reference of producers and users of archival data

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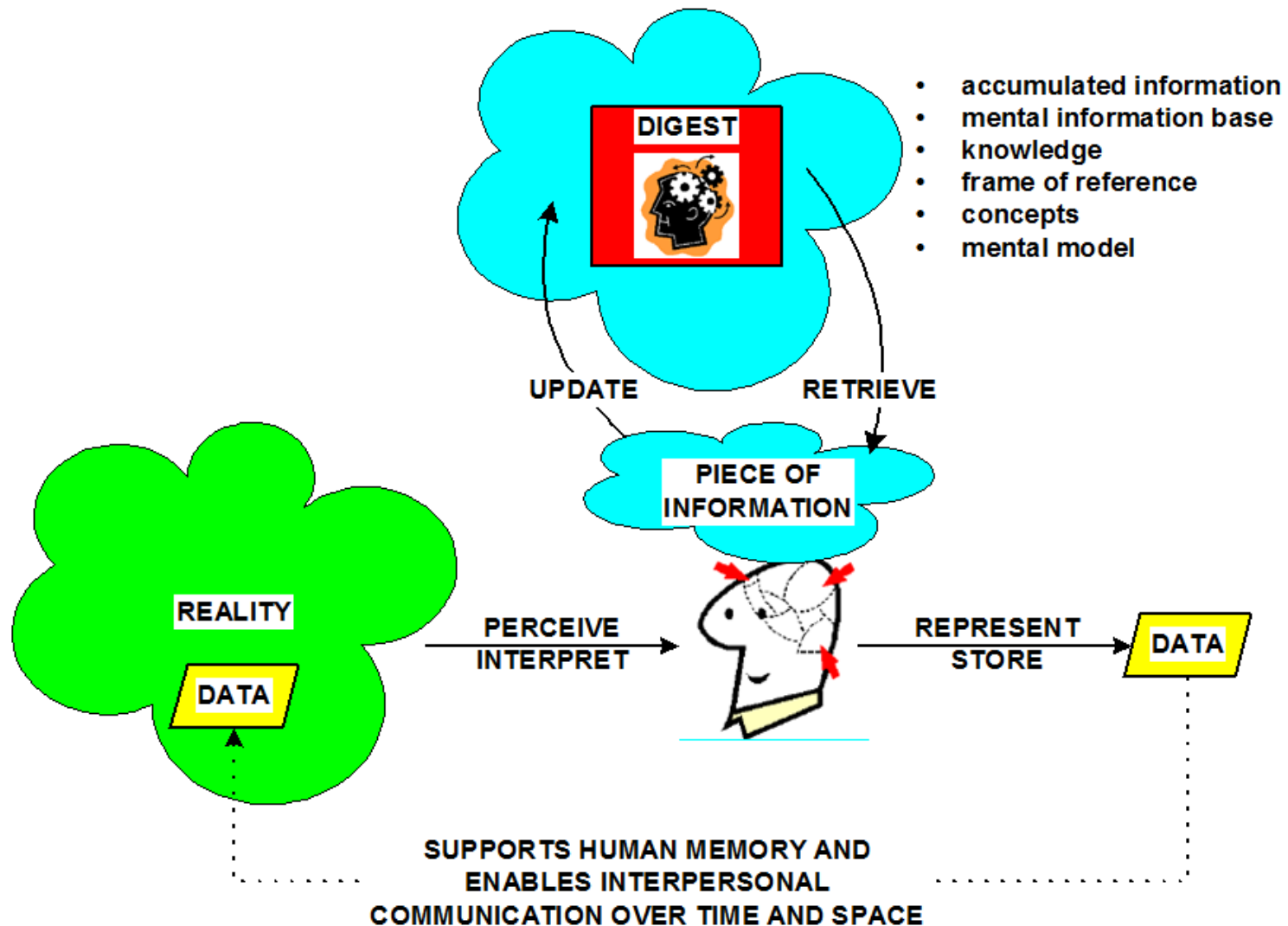
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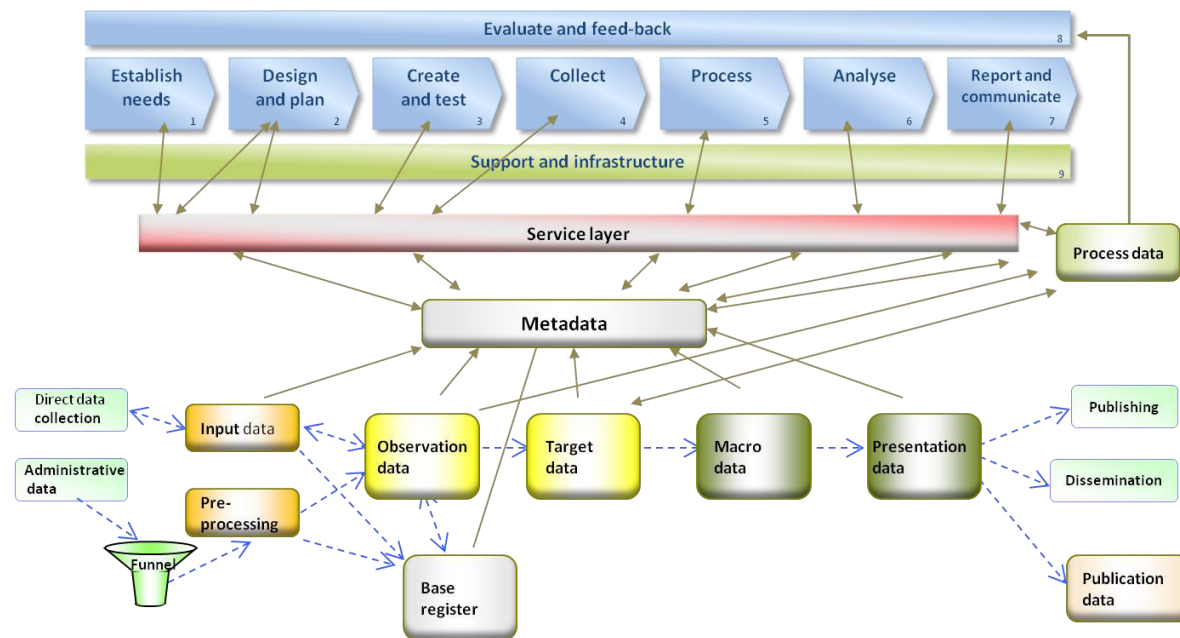
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Reality – Information - Data



Metadata traditions

- The statistical tradition (from 1973 and onwards)
- The library tradition (e.g. Dublin Core)
- The archive tradition (DDI)
- Synthesis: business processes supported by information systems and a corporate data warehouse



The statistical archive system

- Established by Svein Nordbotten in the early 1960's
- Archive-statistical principles:
 - Reuse existing raw data from administrative and statistical sources – for statistical purposes
 - Continuous inflow of data (more or less)
 - Organise data in a systematic way: statistical file system, databases, data warehouse
 - *Ad hoc* production of statistics
 - Systematic descriptions and definitions of data:
 - data and table definition languages; Nordbotten (1967): "[Automatic Files in Statistical Systems](#)"
 - metadata; Sundgren (1973): "[An Infological Approach to Data Bases](#)"
 - Standardised definitions and identifiers enabling flexible integration and combination of data: registers, classifications, standard variables
 - Generalised software
- See also:
 - The Ruggles Report (1965): "[Report of the Committee on the Preservation and Use of Economic Data](#)"
 - EU(2009): "[The production method of EU statistics – a vision for the next decade](#)"

The history of metadata

- Researched and documented by professor **Jane Greenberg**, Director of the Metadata Research Center, University of North Carolina
 - [*Metadata and Digital Information \(2009\)*](#)
- The first known reference to "metadata" appears in **Bo Sundgren** (1973), [*An Infological Approach to Data Bases*](#), pp 104-105
- Claims in the 1990's by **Jack E Myers** to be the originator and owner of the term "metadata" were refuted by the U. S. legal system, with reference to Sundgren (1973) and "the longstanding use of the term in the statistical community".
- In 1986 Myers had registered "Metadata Inc" as a company, and "Metadata" as trade mark of that company. He later started to threaten people and agencies in the U.S. with legal actions, if they did not stop using the term "metadata" as a generic term.
- The **Solicitor of the U.S. Department of the Interior** decided that "Metadata" has entered the public domain by becoming a general term.
- Jack Myers has not been able to provide any documentation supporting his claim to have coined the term "metadata" in the 1960's.

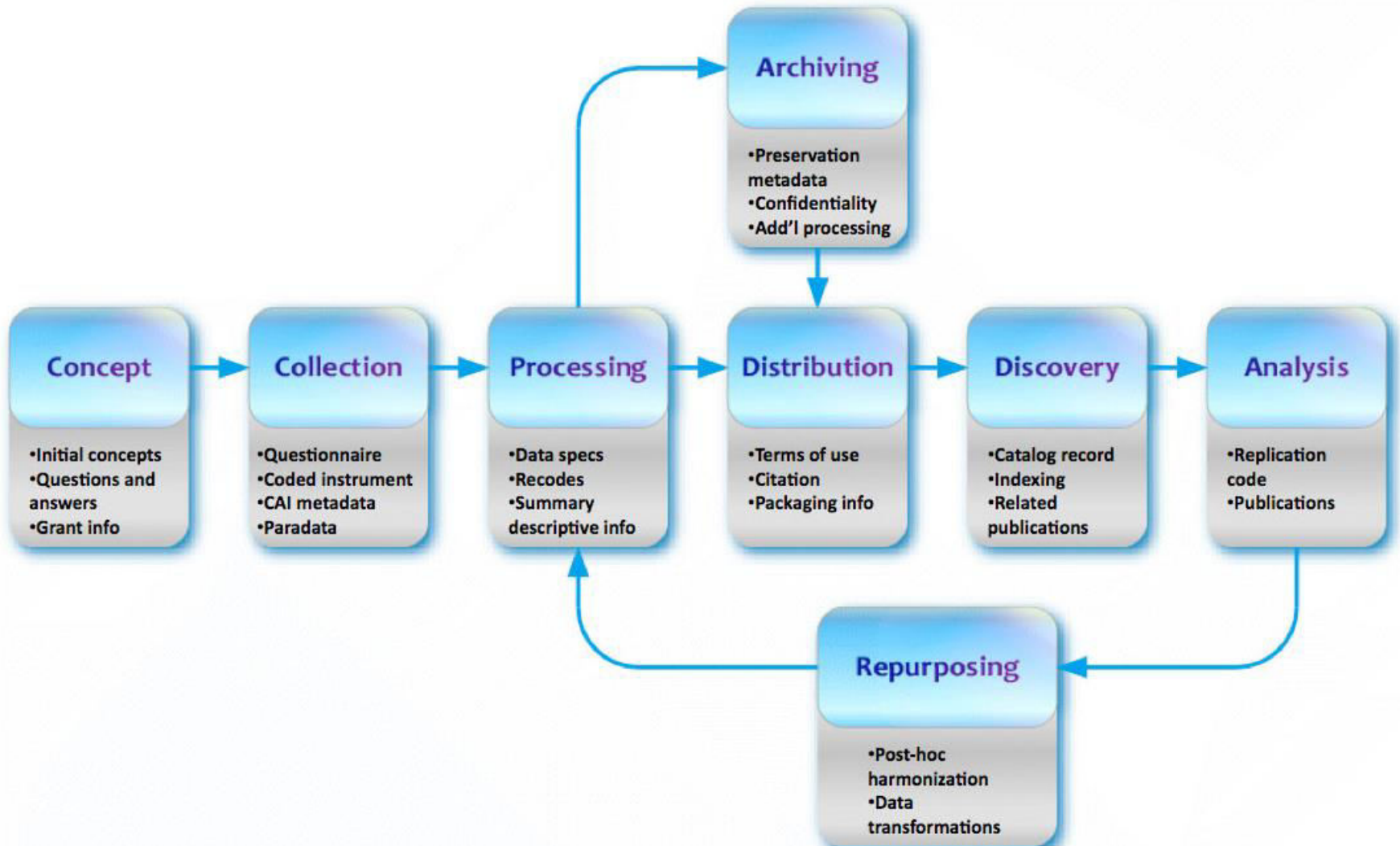
What is DDI? (1)

”A metadata specification for the social and behavioral sciences”
”Document your data across the life cycle”



What is DDI? (2)

“supporting the entire research data life cycle”



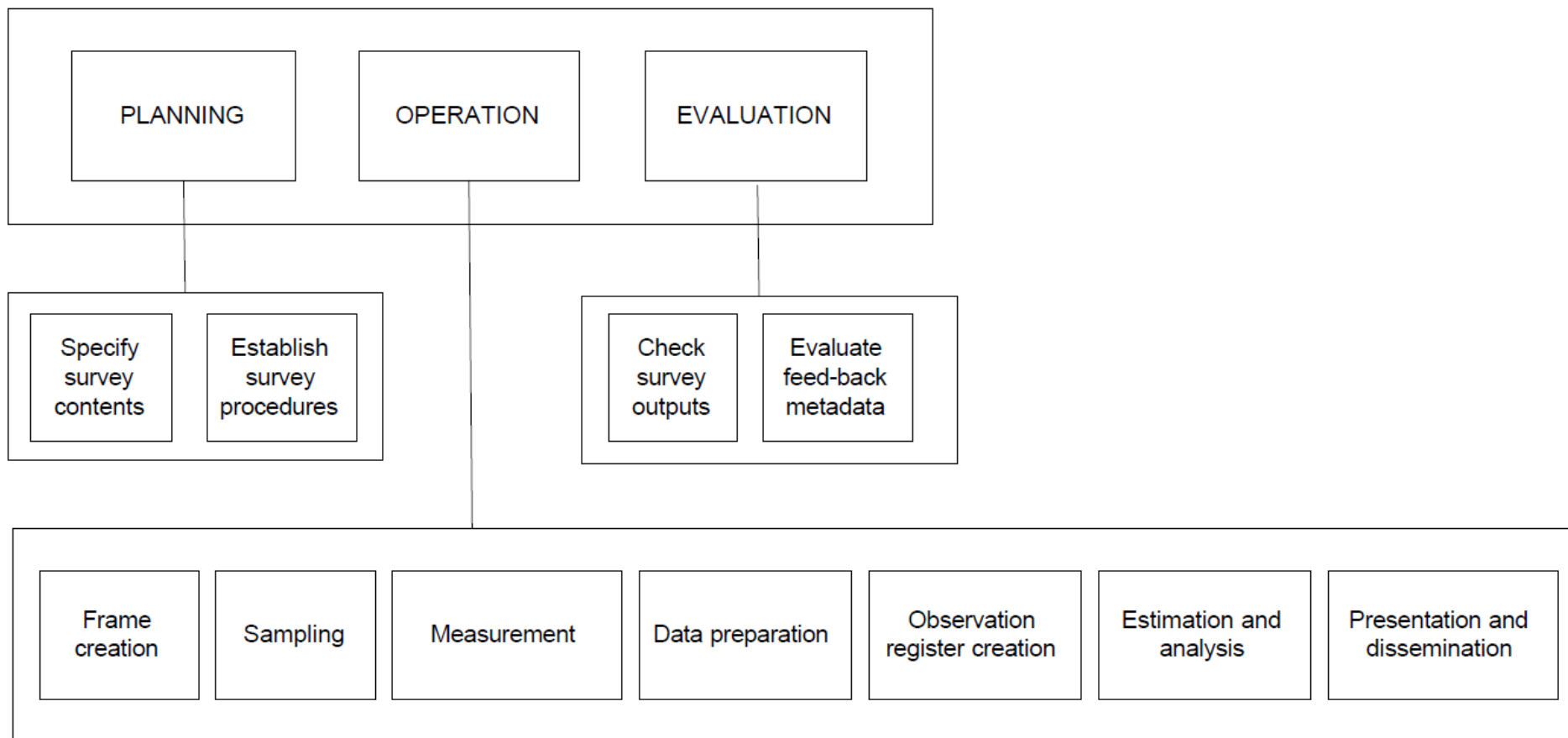
What is DDI? (3)

- **DDI- Codebook** (formerly DDI-2)
 - strictly data oriented
- **DDI- Lifecycle** (formerly DDI-3)
 - process and data oriented

Life cycle models

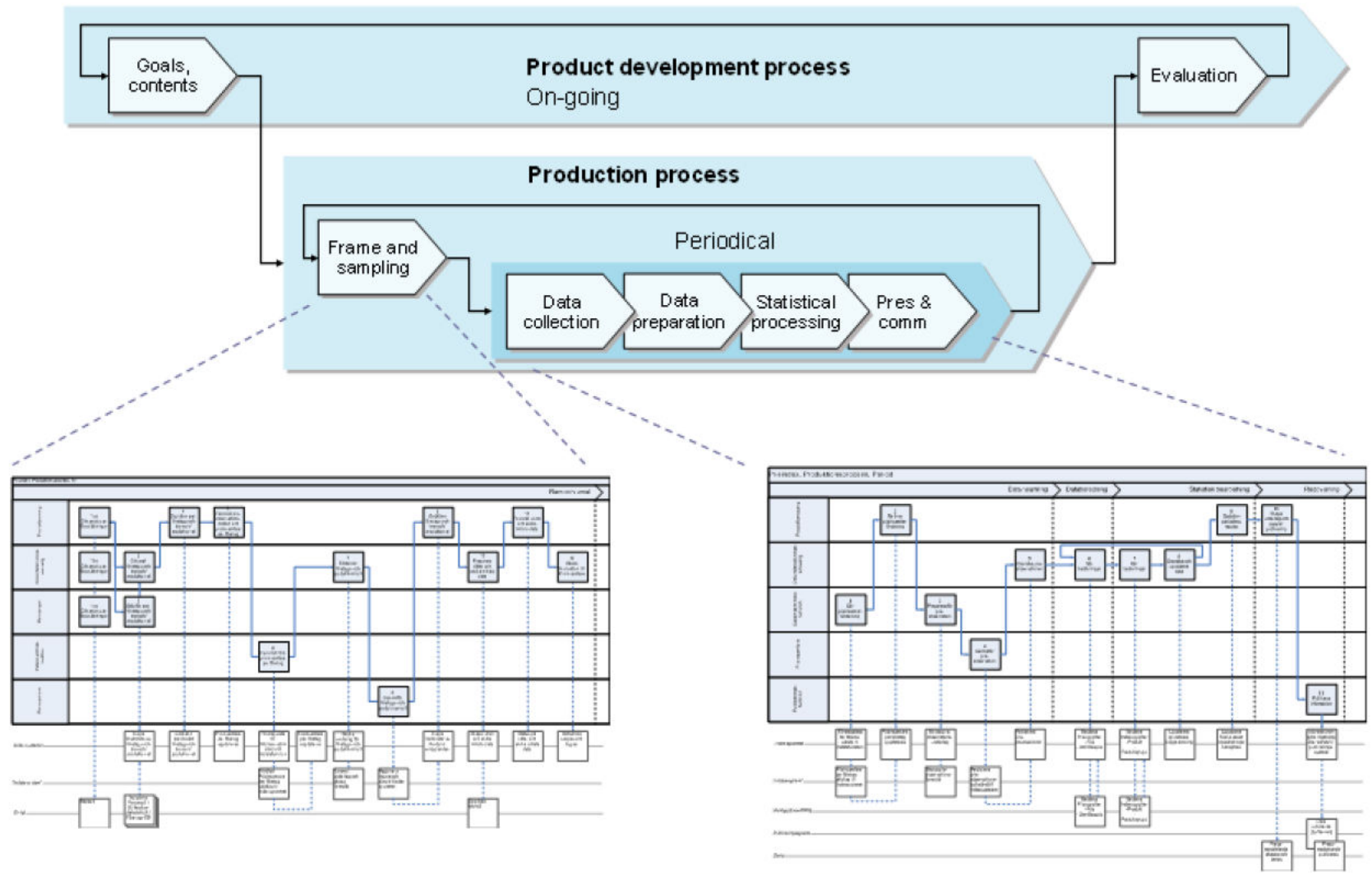
- **Product** life cycle (technical and marketing)
- **Systems development** life cycle (waterfall etc)
- **Software development** life cycle
- **Business process** life cycle
 - **Generic Statistical Business Process Model** (GSBPM)
- **Data/metadata** life cycle
 - **Cycle de vie des données** (CVD), Eurostat model
- **Combined** life cycle
 - **DDI-3**: for the "social science business"

Architecture for statistical systems



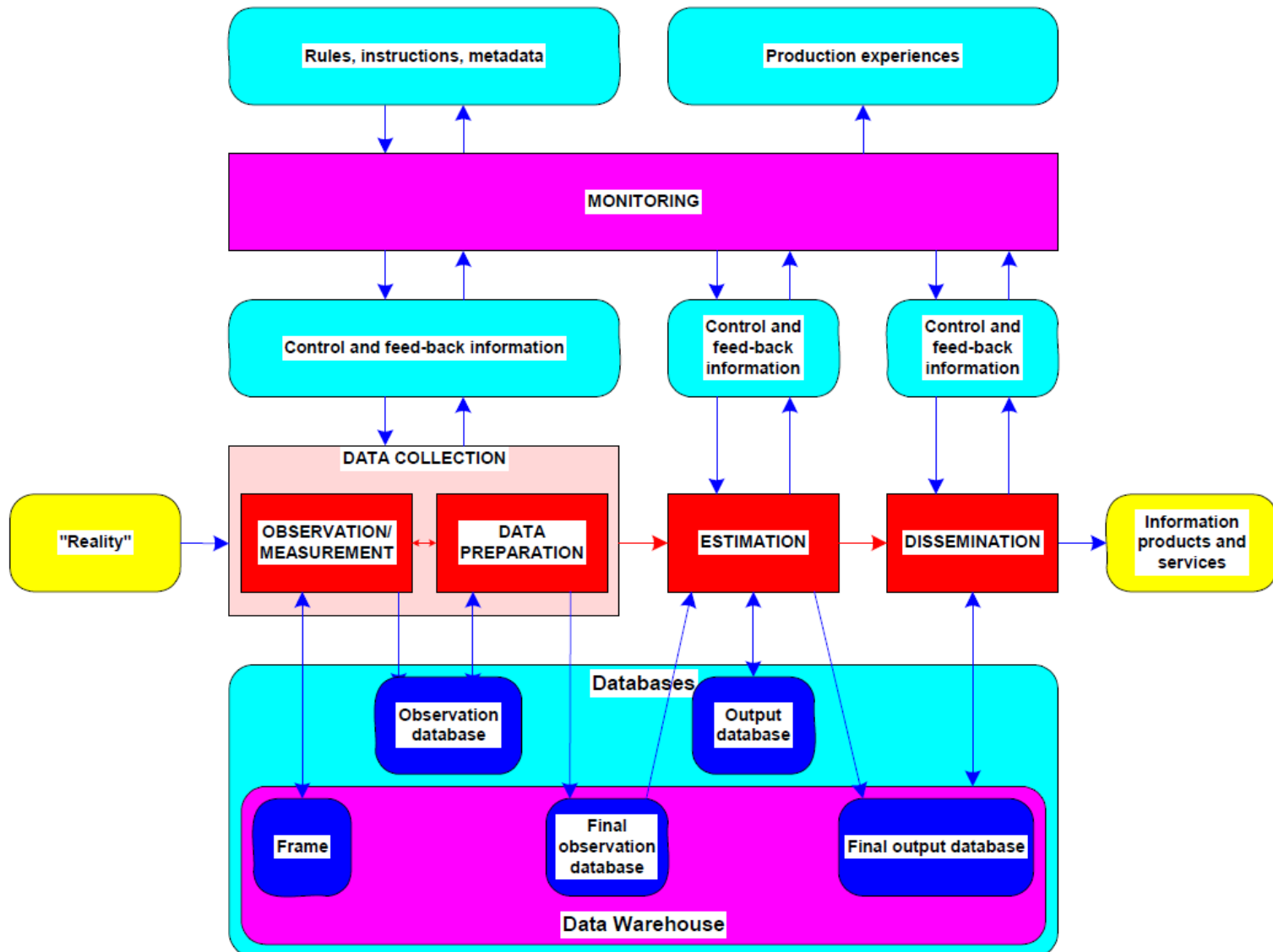
Source: Information Systems Architecture for National and International Statistical Offices – Guidelines and Recommendations, **United Nations, 1999**, http://www.unece.org/stats/documents/information_systems_architecture/1.e.pdf

Statistics production: product development and production processes



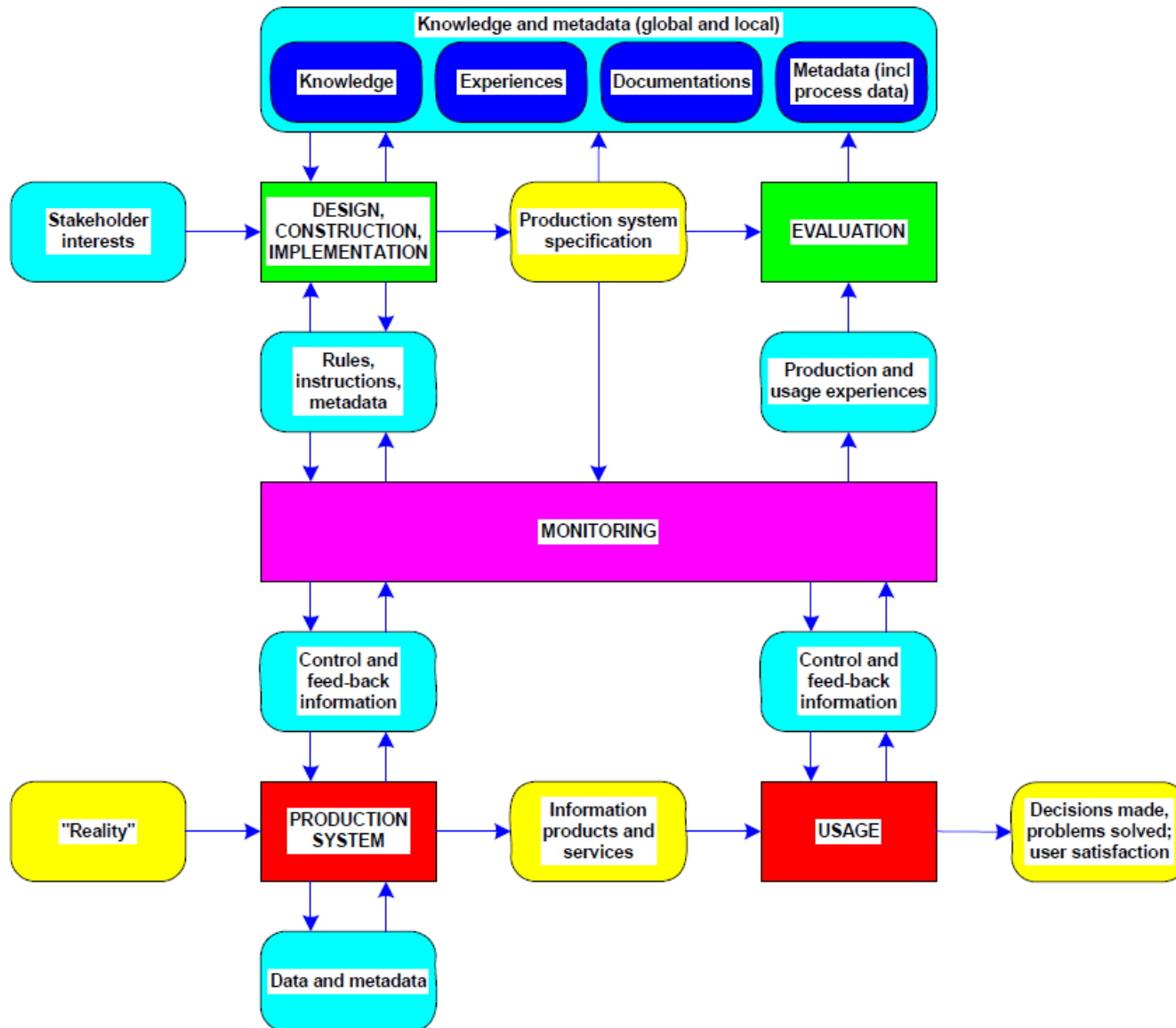
Source: Sundgren (2007) "[Process reengineering at Statistics Sweden](#)", MSIS Geneva.

Basic operations in a database-oriented statistical system



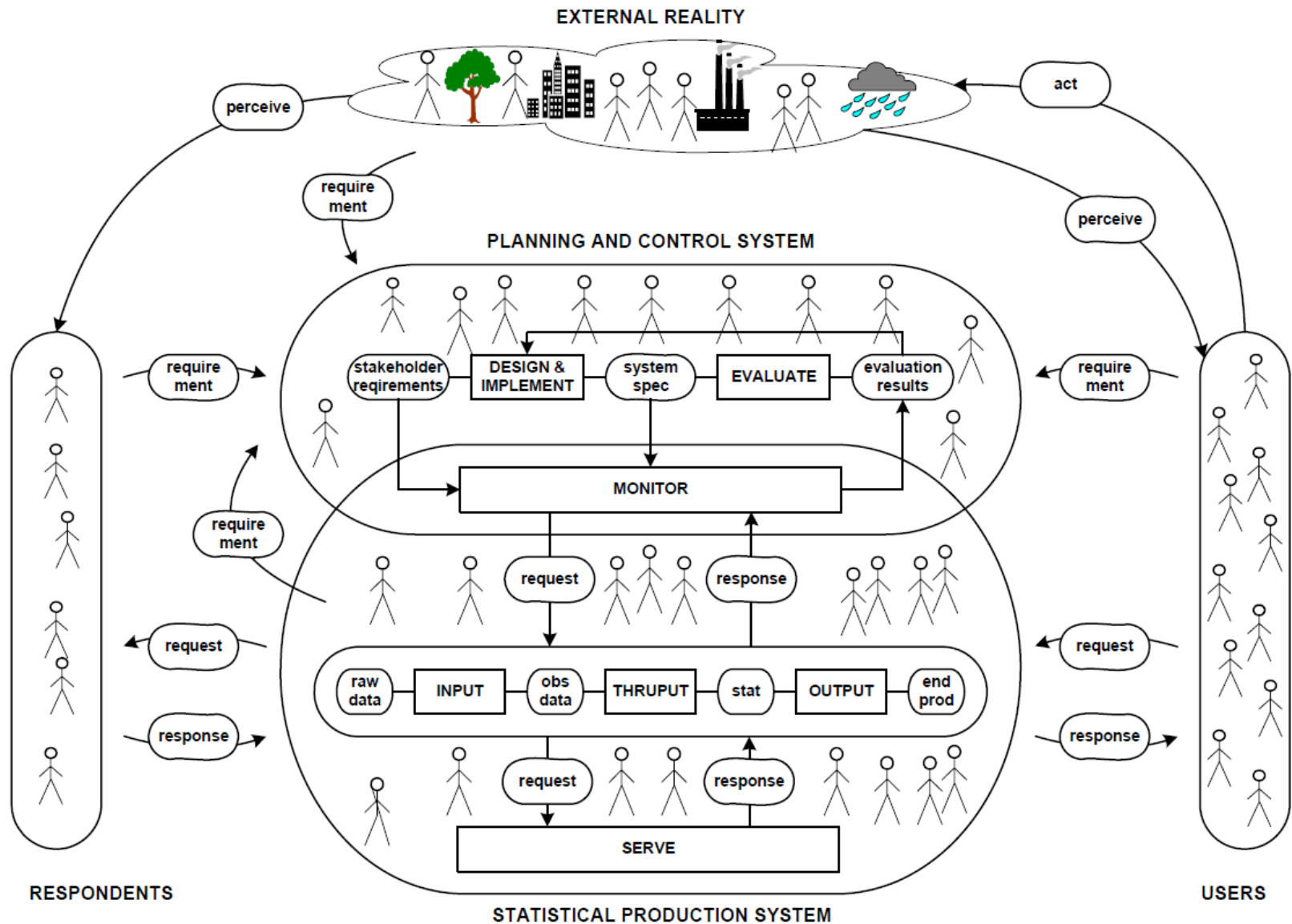
Source: Sundgren (2004b) "[*Statistical systems – some fundamentals*](#)".

Control and execution of a statistical system



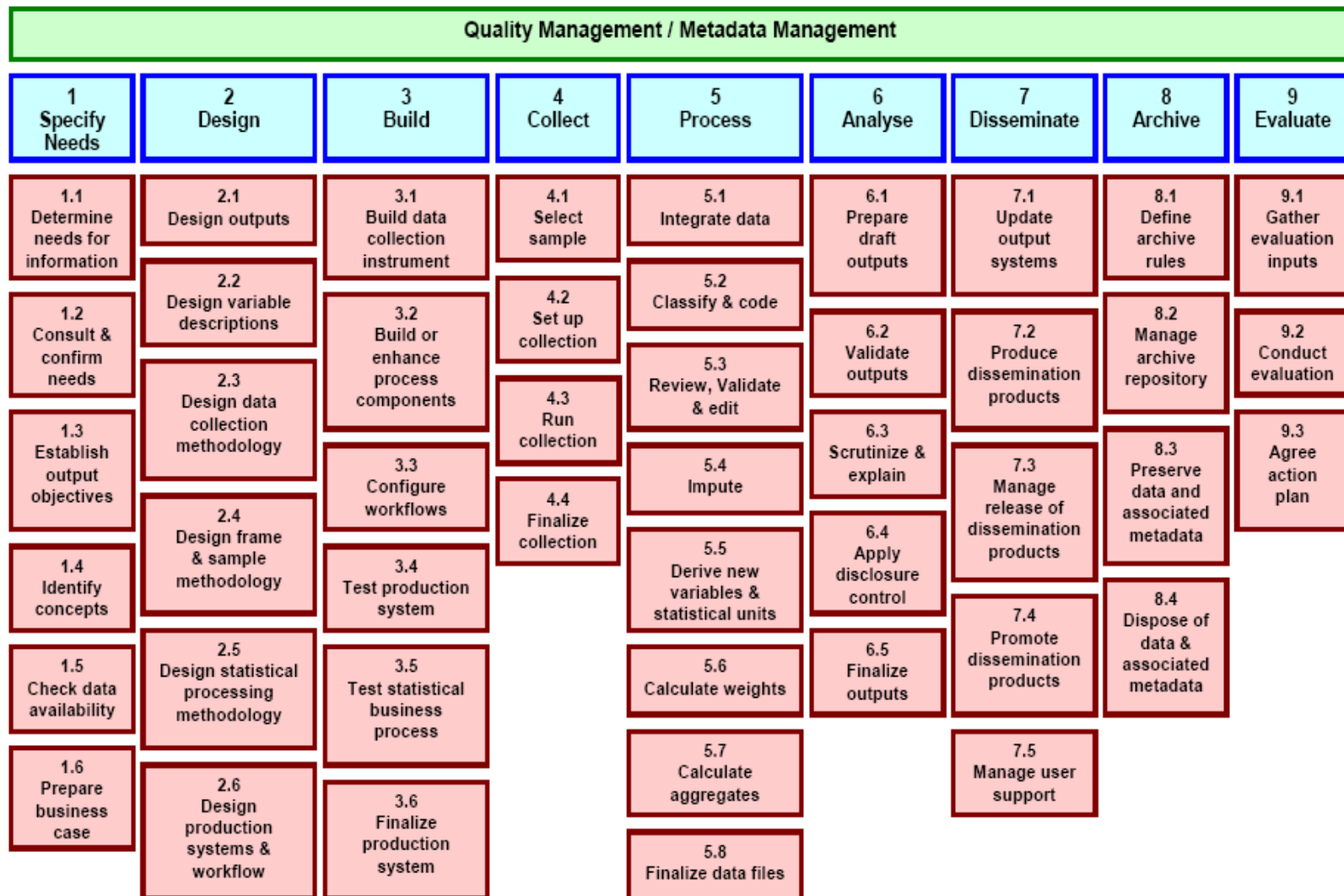
Source: Sundgren (2004b) "[Statistical systems – some fundamentals](#)".

A statistical system and its environment



Source: Sundgren (2004b) "[Statistical systems – some fundamentals](#)".

The Generic Statistical Business Process Model (GSBPM), levels 1 and 2

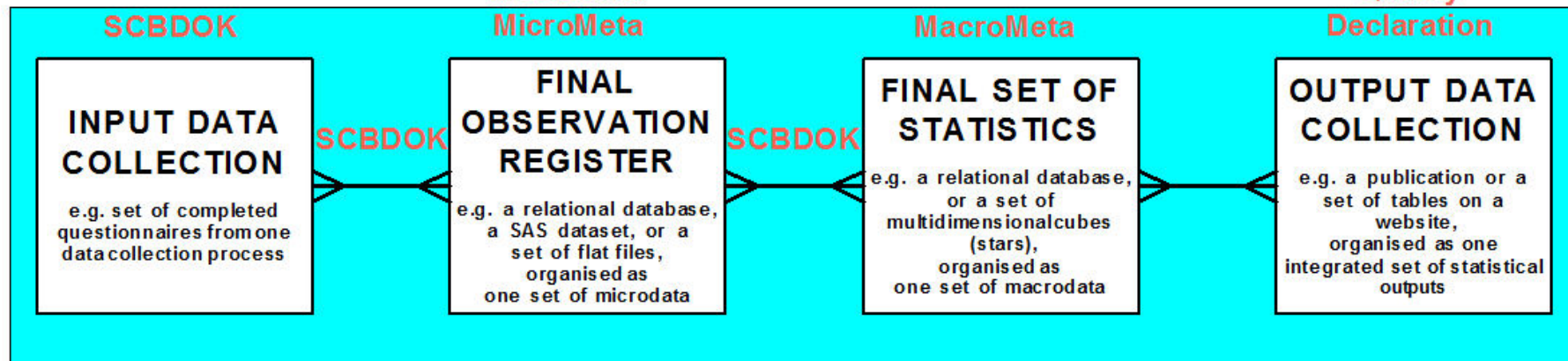
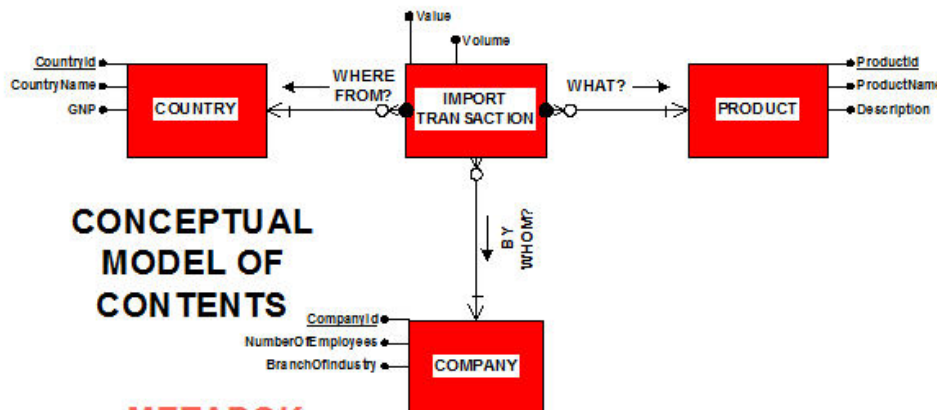


Source: UNECE (2009) "[Generic Statistical Business Process Model](#)". Geneva.

Focusing on data/metadata interfaces

The SCBDOK
conceptual
framework
-data contents
- processes

Classification
Database
(KDB)



Question 1. Name and identity of your company?
(Name of company) Registration number

Question 2. How much (value in €) of different products (identified by product code) has your company imported from different countries (identified by country code) during the last year?
(Use additional table sheets, if needed.)

product code →	product code 1	product code 2	product code 3	product code 4	product code 5
country code 1	value 11	value 12	value 13	value 14	value 15
country code 2	value 21	value 22	value 23	value 24	value 25
country code 3	value 31	value 32	value 33	value 34	value 35

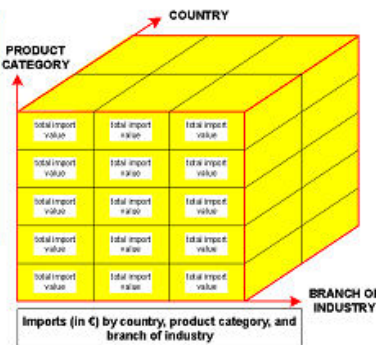
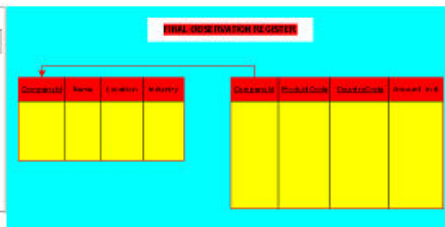


Table 1. Imports (in €) by country, product category, and branch of industry.

		Industry code 1	Industry code 2	Industry code 3	...
Country 1	Product code 1				
	Product code 2				
	...				
Country 2	Product code 1				
	Product code 2				
	...				
Country 3	...				
	...				
	...				

SCBDOK documentation template

SCBDOK 3.0	
0 General information 0.1 Subject matter area 0.2 Statistics area 0.3 Official statistics? 0.4 Responsibility 0.5 Producer 0.6 Mandatory response? 0.7 Secrecy 0.8 Destruction rules 0.9 EU regulation 0.10 Purpose and history 0.11 Users and usage 0.12 General approach to implementation 0.13 Planned changes	1 Contents overview 1.1 Observation characteristics 1.2 Statistical target characteristics 1.3 Outputs: microdata and statistics 1.4 Documentation and metadata 2 Data collection 2.1 Frame and frame procedure 2.2 Sampling procedure (if applicable) 2.3 Measurement instruments 2.4 Data collection procedure 2.5 Data preparation
3 Final observation registers 3.1 Production versions 3.2 Archive versions 3.3 Experiences from the latest collection round	4 Statistical processing and presentation 4.1 Estimations: assumptions and formulas 4.2 Presentation and dissemination procedures
5 Data processing system	6 Logbook

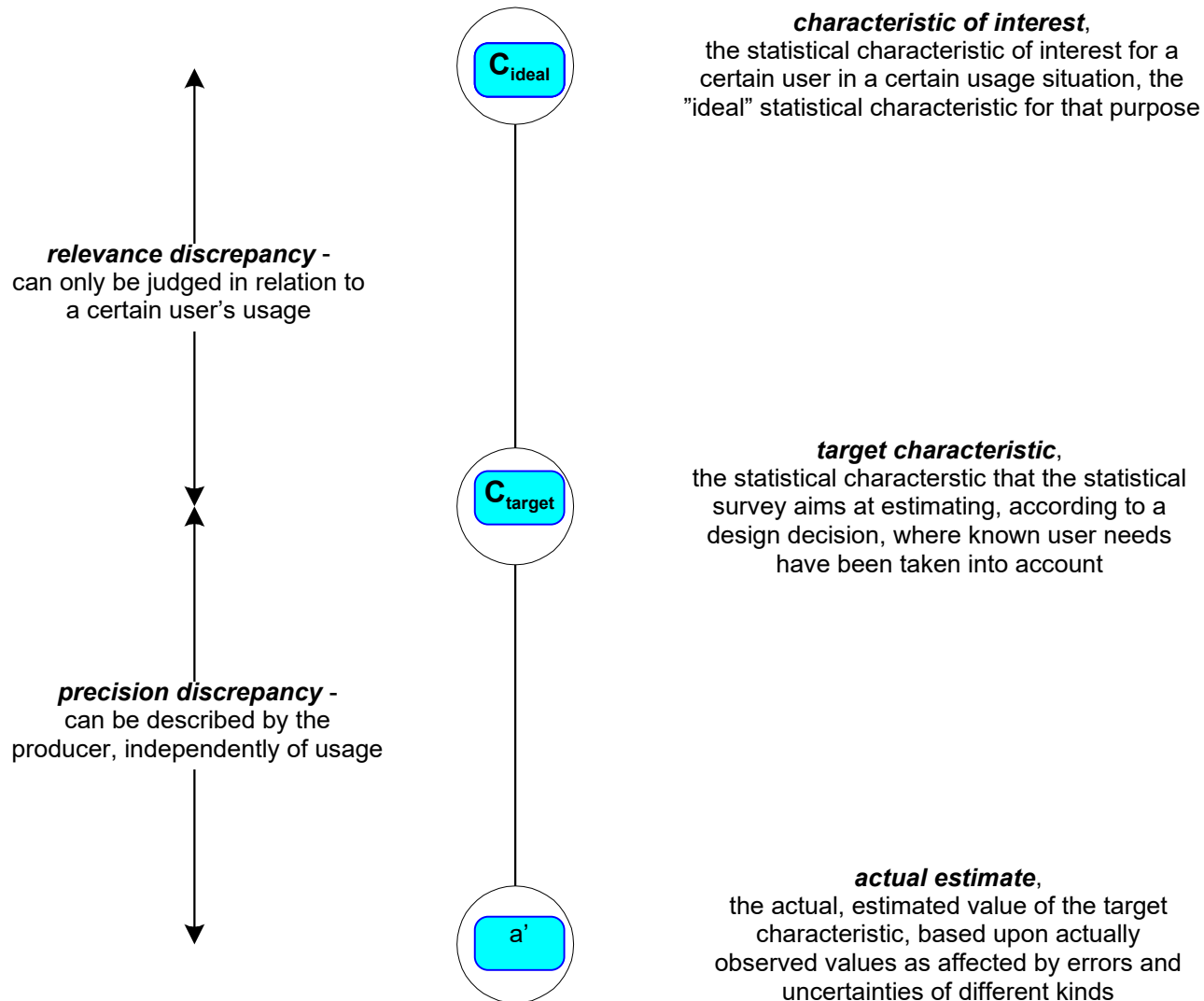
Source: [Sundgren \(2001\) "Documentation and quality in official statistics"](#).

Quality Declaration Template

Quality Declaration Template	
1 Contents 1.1 Statistical target characteristics 1.1.1 Objects ³ and population 1.1.2 Variables 1.1.3 Statistical measures 1.1.4 Study domains 1.1.5 Reference time 1.2 Comprehensiveness	2 Accuracy 2.1 Overall accuracy 2.2 Sources of inaccuracy ⁴ 2.2.1 Sampling 2.2.2 Coverage 2.2.3 Measurement 2.2.4 Non-response 2.2.5 Data processing ⁵ 2.2.6 Model assumptions 2.3 Presentation of accuracy measures
3 Timeliness 3.1 Frequency 3.2 Production time 3.3 Punctuality	4 Coherence especially comparability 4.1 Comparability over time 4.2 Comparability over space 4.3 Coherence in general
5 Availability and clarity 5.1 Forms of dissemination 5.2 Presentation 5.3 Documentation 5.4 Access to microdata 5.5 Information services	

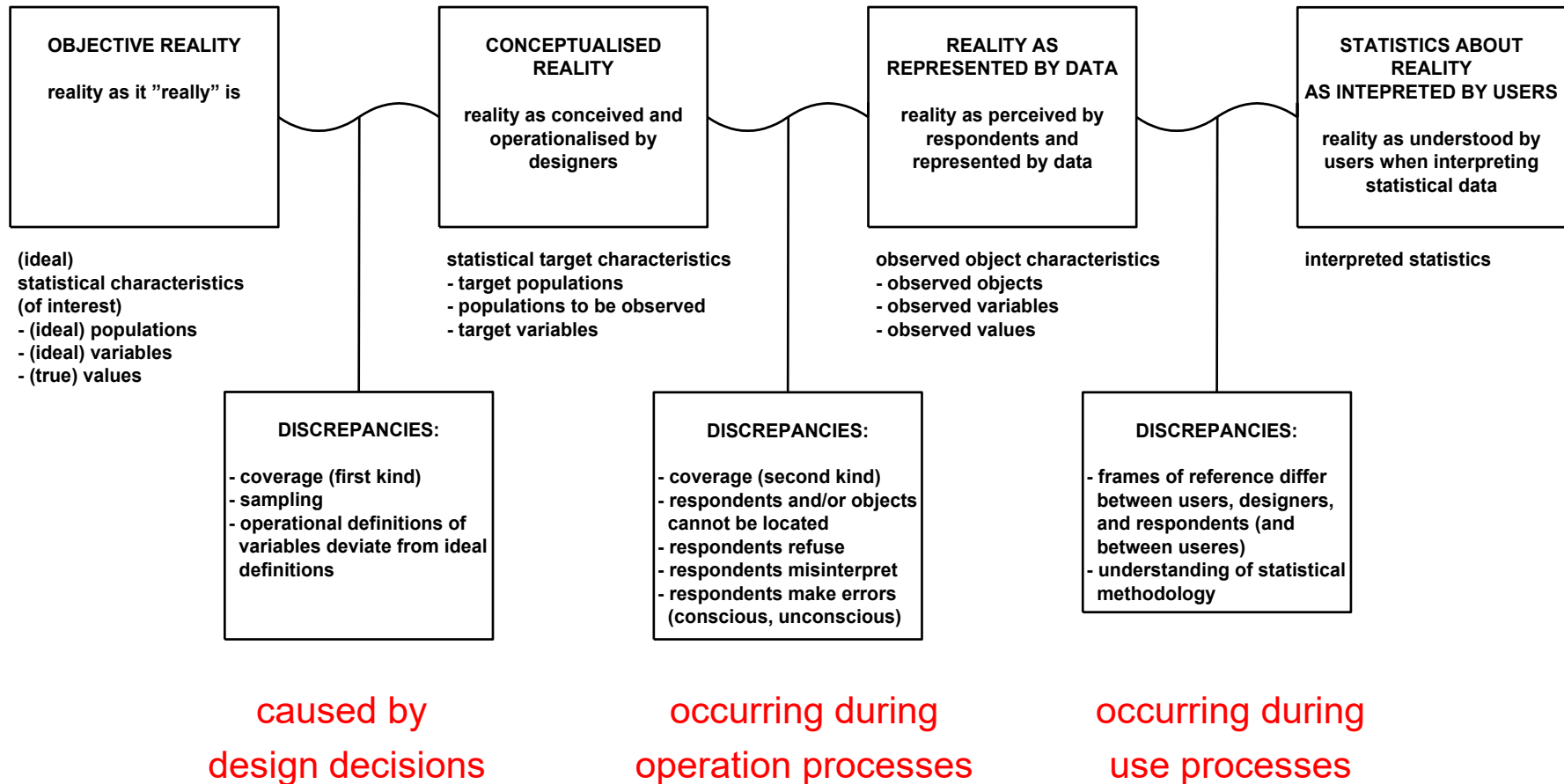
Source: [Sundgren \(2001\) "Documentation and quality in official statistics"](#).

The quality of statistical data as affected by different discrepancies



Source: Sundgren (1995) [Guidelines for the Modelling of Statistical Data and Metadata](#).

Discrepancies between reality “as it is” and as it is reflected by statistics: which they are, and why and where they occur



Source: Sundgren (2004b) [“Statistical systems – some fundamentals”](#).

Statistical characteristic

- a **statistical measure** (m) applied on
- the (true) **values** of a **variable** (V); V may be a vector
- for the **objects** in a **population** (O)
- O.V.m = statistical characteristic
- O.V = object characteristic
- V.m = **parameter**

Examples of statistical characteristics

- number of persons living in Sweden at the end of 2001
- average income of persons living in Sweden at the end of 2001
- correlation between sex and income for persons living in Sweden at the end of 2001

Statistic

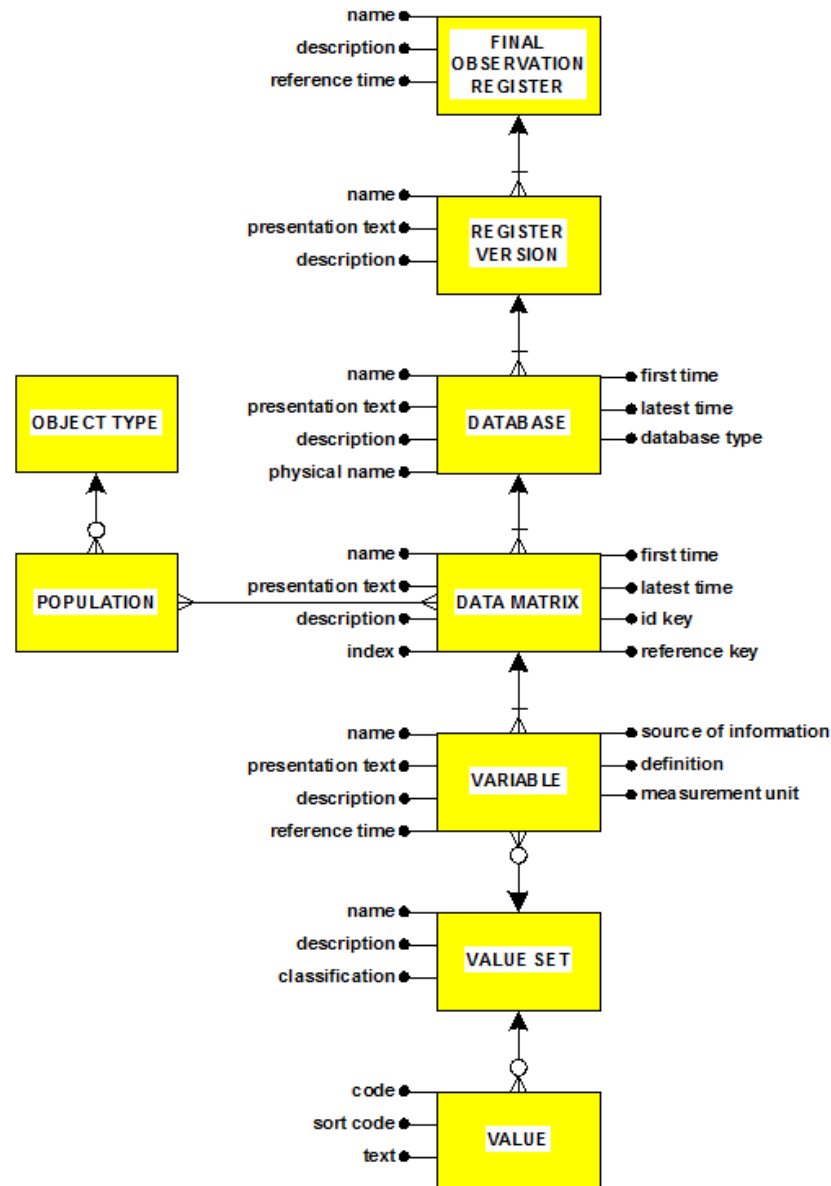
- an **estimator** (e) applied on
- **observed values** of an **observed variable** (V');
- for a set of **observed objects** (O') **allegedly** belonging to a **population** (O)

Ideally the value of a statistic $O'.V'.e$ should be "close to the true value of the statistical characteristic $O.V.m$ that it aims at estimating"

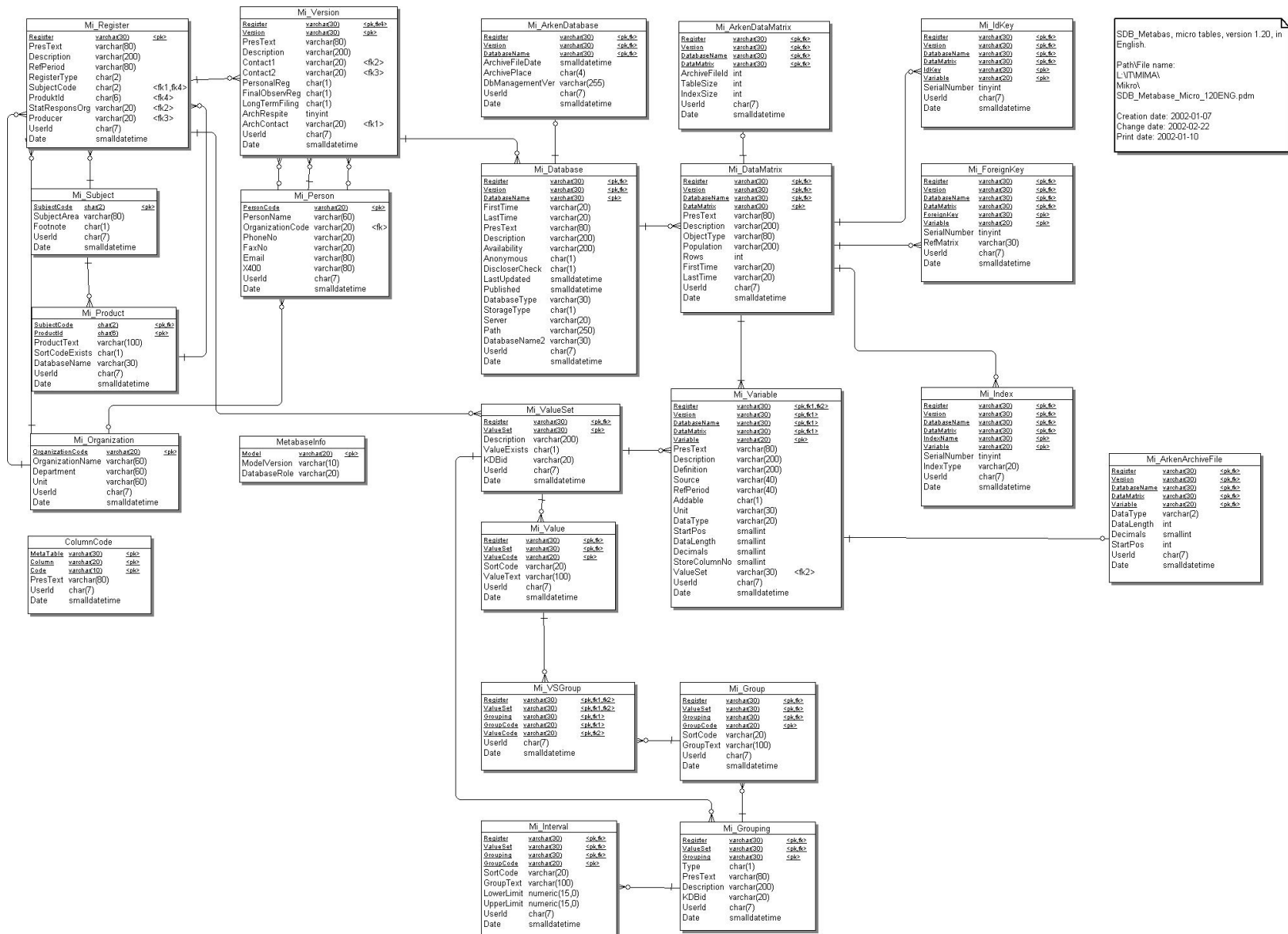
Examples

- the estimated number of persons living in Sweden at the end of 2001
- the estimated average income of persons living in Sweden at the end of 2001
- the estimated correlation between sex and income for persons living in Sweden at the end of 2001

Metadata overview for final observation registers

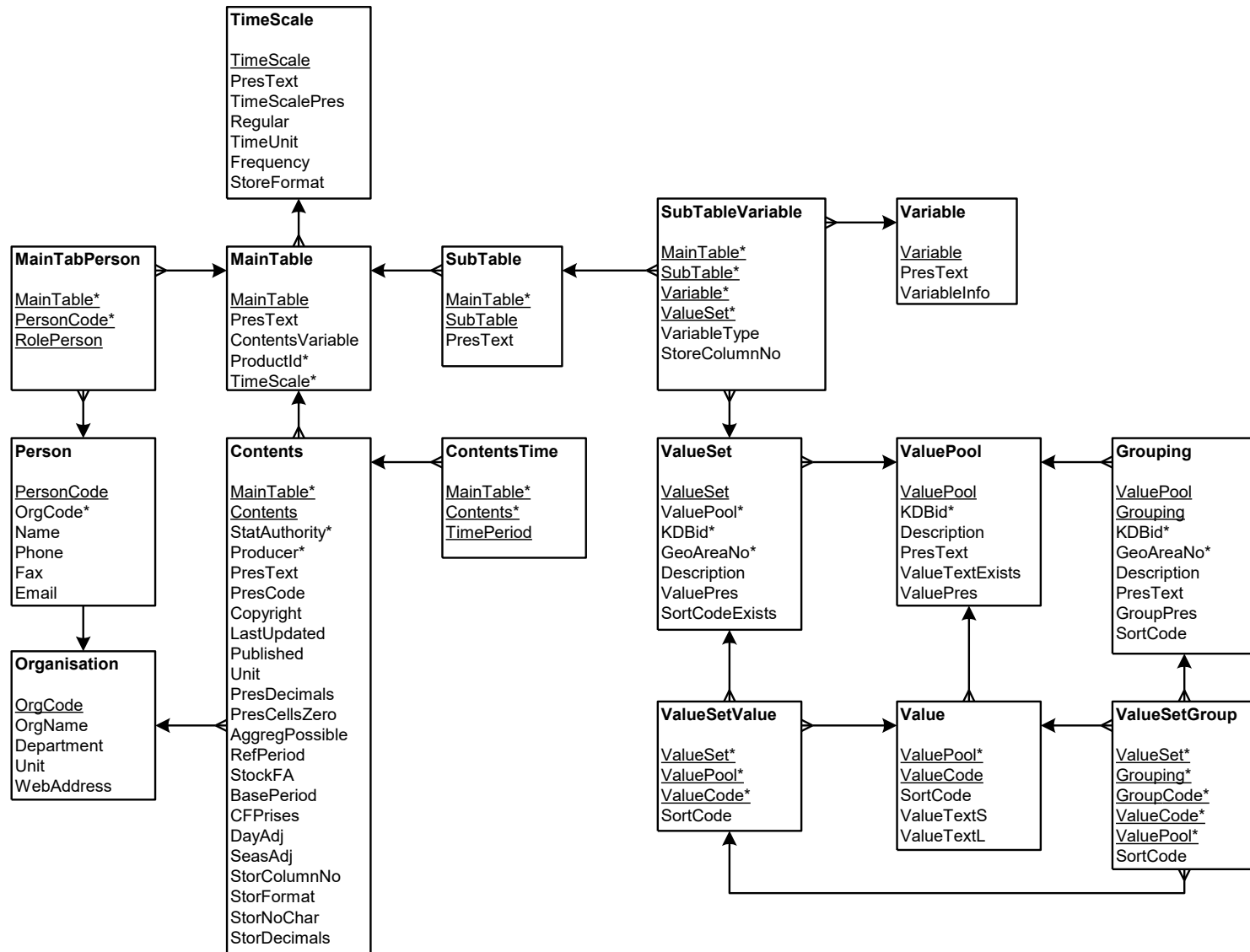


Complete MicroMeta model

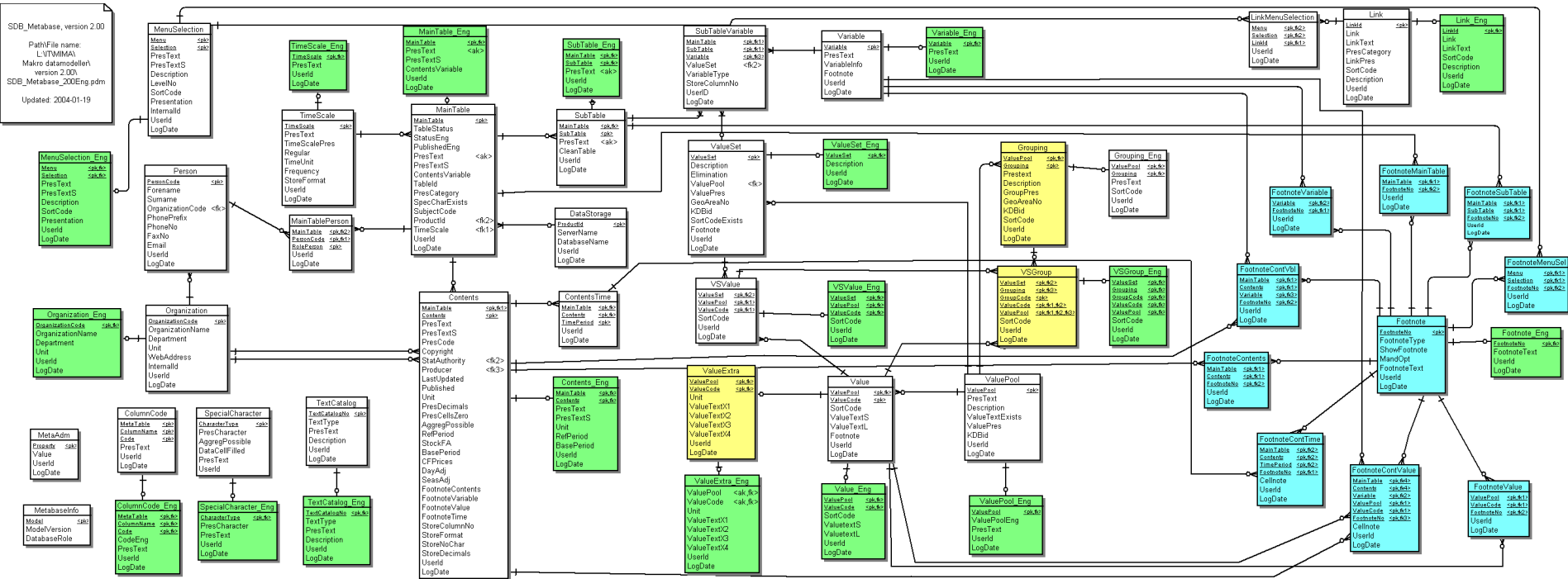


Source: Sundgren&Lindblom (2004) *"The metadata system at Statistics Sweden in an international perspective"*

MacroMeta: simplified overview



MacroMeta: complete model



Source: Sundgren&Lindblom (2004) "[The metadata system at Statistics Sweden in an international perspective](#)"

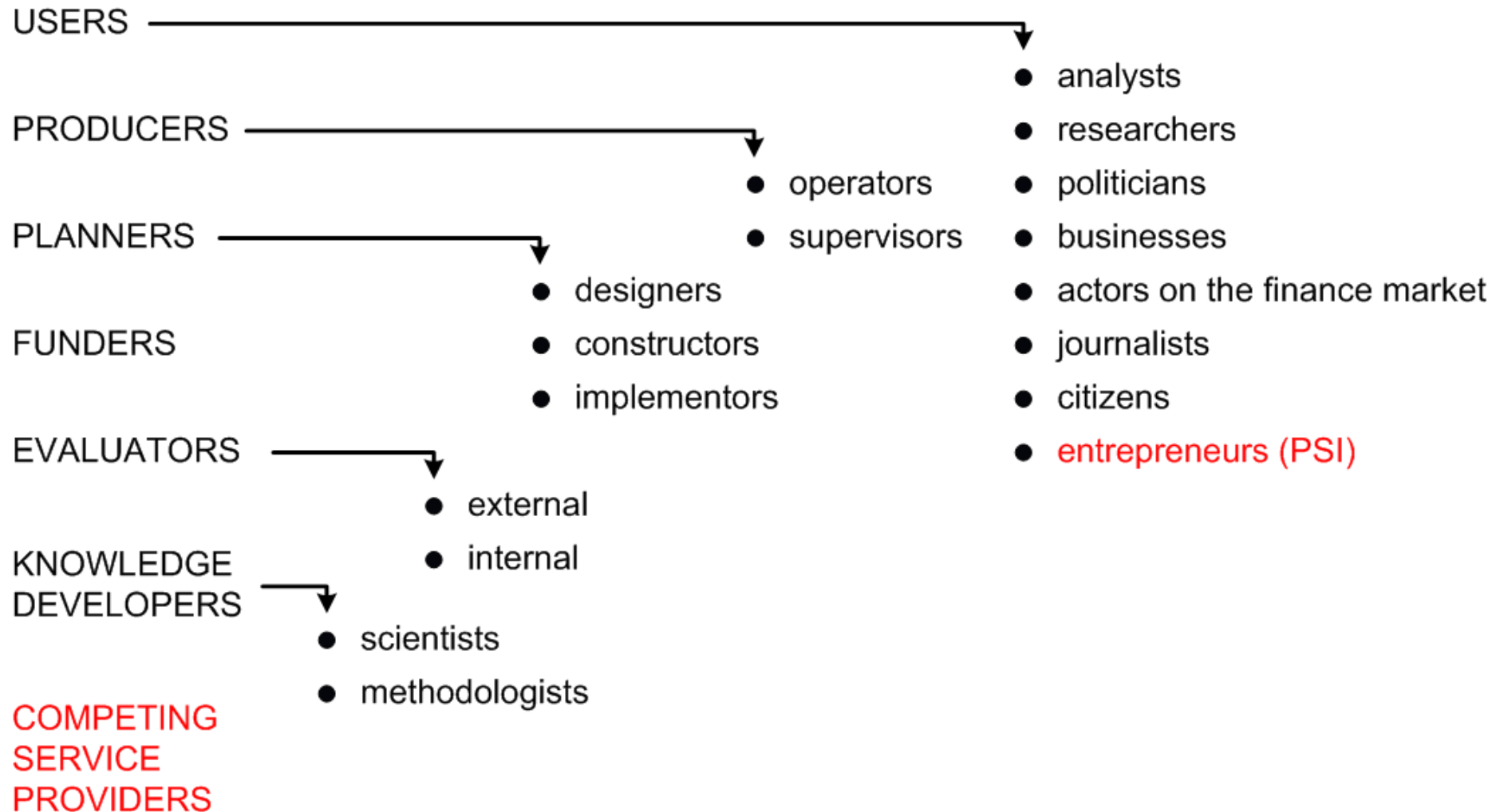
Three critical success factors for documentation and metadata

- **Motivation:** How to respond to the arguments against documentation-related work?
- **Contents:** Which metadata are needed by which stakeholders for which purposes?
- **Management:** How to manage a metadata system in an efficient and sustainable way?

Arguments against documentation

- Time and costs: "we don't have the resources"
- "We have more important things to do"
- Dull, not fun, not rewarding
- Competent key persons are scarce resources
- "I know everything – come to me and ask"
- "I don't want to lose my knowledge monopoly"
- "The users don't ask for documentation"
- Easy-to-use tools are not available
- Documentation is produced as a separate activity
- There are too many types of documentation, contents overlapping and not well motivated, duplication of work

Stakeholders in official statistics and in metadata about official statistics



Documentation/metadata objects

- **Datasets**

- information contents
 - object types and populations
 - object relations and relational objects
 - variables and value sets
- physical datasets

- **Processes and systems**

- **Instruments and tools**

- methods, algorithms, programs
- questionnaires and other measurement instruments
- registers, classifications, other auxiliary datasets
- metadata, documentation

Documentation/metadata variables

- **For datasets:**
 - definitions, verbal or formal
 - quality variables, by quality component
 - technical metadata, e.g. storage format
- **For processes and systems:**
 - references to input and output datasets
 - references to instruments and tools
 - process data (paradata) generated by the processes
- **For instruments and tools:**
 - documentation of instruments and tools
 - the instruments and tools themselves (*in extenso*)
 - references to systems and processes using them
 - user experiences

Use, production, and recycling of documentation and metadata

DOCUMENTATION/METADATA INPUTS

- Stakeholder requirements ●
- General knowledge ●
- Experiences ●
- Recirculated process data ●
- Information about available data&metadata resources ●
- Information about available tools and instruments ●

(RE) PLANNING

- design
- construction
- implementation

DOCUMENTATION/METADATA OUTPUTS

- Planning decisions
- System specification and documentation
- Test results (process data)
- Detailed instructions for the execution
- Tutorials and user-oriented instructions
- Costs (process data)

- System specification and documentation ●
- Metadata driving the processes ●
- Recirculated process data ●
- Detailed instructions for the execution ●

EXECUTION

- operation
- monitoring

- Metadata created and transformed by the processes
- Process data generated by the processes
- Updated system documentation
- Costs (process data)

- User-oriented documentation, tutorials, and instructions ●
- Metadata driving use processes ●
- Recirculated process data ●

USE

- professional
- interested citizens

- Metadata created and transformed by use processes
- Process data generated by use processes
- Documented user experiences and proposals

- Stakeholder requirements ●
- General knowledge ●
- Experiences ●
- Complete system documentation ●
- Recirculated process data ●
- Documented user experiences and proposals ●

EVALUATION

- internal
- external

- Documented replanning proposals

The Swedish Statistics Commission

- Set up by the Swedish government to evaluate and improve Statistics Sweden and the Swedish Statistical System
- Principal investigator: Bengt Westerberg, a former minister of social affairs and leader of the Swedish Liberal Party
- 10 experts
- Final report to be delivered in December 2012

Some tasks of the Statistics Commission

- Analyse centralised vs decentralised system
- Examine in particular the quality and accessibility of statistics, including documentation, pricing, and confidentiality
- Analyse what it means for statistics production that state authorities as a rule should not sell goods and services on the market
- Analyse the impact on SCB of the PSI Act
- Propose measures to ensure and improve quality, accessibility, and documentation, including a monitoring system
- Propose a strengthening of SCB's cooperation with universities and other agencies
- Propose how the system of official statistics should be designed
- Propose changes in Swedish regulations as the result of new regulations and expected changes on the European level, e.g. the Code of Practice for European Statistics, the PSI Directive, and "the EU vision for official statistics" (see enclosed slide)
- Propose any constitutional amendments deemed necessary

Summary of my proposals concerning quality, availability, and documentation

- Independent monitoring system
- Free access to methods and tools (software, databases, registers, metadata, documentation, etc) that have been developed for official statistics and funded by public money
- Continuously ongoing development of knowledge, methods, and general tools, funded by public money
- Systematic implementation of best practices

Summary of the EU vision for official statistics

- **Current situation: the augmented stovepipe model**
 - respondents asked for the same information more than once
 - not adapted to collect data across domains
 - little standardisation and coordination between areas
- **Demands for change:**
 - new information needs, often across domains, often ad hoc (e.g. in crises)
 - decrease response burden
 - use new ICT methods and tools to increase efficiency
- **Consequences on the level of Member States:**
 - holistic approach, stovepipes replaced by integrated production systems around a data warehouse
 - data obtained from existing administrative data and/or extracted directly from company accounts, combining survey data with administrative data, new efforts to ensure the quality of the data
- **Consequences on the EU level:**
 - Horizontal integration similar to the Member State level
 - Two elements of vertical integration: (i) collaborative networks, and
 - (ii) direct production for the EU level, when there is no need for national data

Golden rules for metadata systems

- Experiences of metadata systems from Sweden and elsewhere have been summarized in a set of “golden rules”, aiming at designers, project managers/co-coordinators, and top managers, respectively.
- The rules are formulated and elaborated in:
 - Sundgren (2003a) “[Developing and implementing statistical metainformation systems](#)”, Deliverable from EU project
 - Sundgren (2003b) “[Strategies for development and implementation of statistical metadata systems](#)”, ISI Berlin
 - Sundgren & Lindblom (2004) “[The metadata system at Statistics Sweden in an international perspective](#)”, Prague
 - Sundgren (2004) “[Metadata systems in statistical production processes – For which purposes are they needed, and how can they best be organised?](#)”, UNECE/Eurostat/OECD, Geneva

Golden rules (1):

If you are a designer...

- Make metadata-related work an integrated part of the business processes of the organisation.
- Capture metadata at their natural sources, preferably as by-products of other processes.
- Never capture the same metadata twice.
- Avoid un-coordinated capturing of similar metadata – build value chains instead.
- Whenever a new metadata need occurs, try to satisfy it by using and transforming existing metadata, possibly enriched by some additional, non-redundant metadata input.
- Transform data and accompanying metadata in synchronised, parallel processes, fully automated whenever possible.
- Do not forget that metadata have to be updated and maintained, and that old versions may often have to be preserved.

Golden rules 2:

If you are the project co-ordinator...

- Make sure that there are clearly identified “customers” for all metadata processes, and that all metadata capturing will create value for stakeholders.
- Form coalitions around metadata projects.
- Make sure that top management is committed. Most metadata projects are dependent on constructive co-operation from all parts of the organisation.
- Organise the metadata project in such a way that it brings about concrete and useful results at regular and frequent intervals.

Golden rules (3):

If you are the top manager...

- Make sure that your organisation has a metadata strategy, including a global architecture and an implementation plan, and check how proposed metadata projects fit into the strategy.
- Either commit yourself to a metadata project – or don't let it happen. Lukewarm enthusiasm is the last thing a metadata project needs.
- If a metadata project should go wrong – cancel it; don't throw good money after bad money.
- When a metadata project fails, make a diagnosis, learn from the mistakes, and do it better next time.
- Make sure that your organisation also learns from failures and successes in other statistical organisations.
- Make systematic use of metadata systems for capturing and organising tacit knowledge of individual persons in order to make it available to the organisation as a whole and to external users of statistics.

Thank you for your attention!